Classification and review of MTPA algorithms for synchronous reluctance and interior permanent magnet motor drives

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Abstract

The MTPA algorithms are attractive solutions for reaching very high energy efficient drives. The paper presents a general formulation of the MTPA problem statement, including also the magnetic saturation in the formulation. A classification and a review of the MTPA techniques is proposed in this paper. A distinctive feature of the proposed classification is the highlighting of the most important characteristics (e.g. needs of motor parameters, signal injection, self-commissioning tests, etc.) of each technique. Also, offline and online techniques are thoroughly discussed. All synchronous motors, with or without permanent magnets, are considered.

I. Introduction

Starting with the massive spread of variable speed drives (VSD) in the market in the '80s, the minimisation of the energy consumption [1]–[3] was one of the engineers' most prominent concerns. Nowadays, the preponderance of VSD is implementing synchronous motors equipped with permanent magnets and with an isotropic magnetic structure. During the 90's, there was an active research about maximising both efficiency and range of speed [4]–[6]. During the last decade, there has been an increasing interest towards synchronous motors with anisotropic rotor structures, with or without permanent magnets. They present a double torque generation mechanism (PM-based and reluctance), which allows to reduce or exclude the amount of permanent magnet material. Essentially, as much as rare-earth materials become a cost-critical commodity, new constructive topologies appears.

Elaborated control techniques aiming at exploiting the double torque generation mechanism in the most efficient way are, thus, of great importance. These techniques are called *maximum torque-per-ampere* (MTPA) and it is often used as a synonym of *maximum efficiency*, since copper losses are often predominant compared to iron losses in synchronous motors. By the way, the evaluation of the iron losses is not a trivial task and it is often neglected [7]. It is worth mentioning that, sometimes, the MTPA techniques are also called *maximum torque-per-current* (MTPC) [8], which has the advantage of not using the unit of measure (Ampere) to indicate the variable that has to be minimised.

In principle, the MTPA methods can be divided among online and offline techniques. The definition and classification of MTPA techniques present in literature is not trivial, and it is one of the outputs of this paper. The online techniques are expected to work only using information which are gathered during the normal operation of the motor. The offline ones reach the MTPA condition by means of information obtained prior to the normal operation of the motor.

The massive research production on MTPA techniques is, thus, blurred by the lacking of a clear MTPA techniques classification. Furthermore, many proposed techniques lacks of the necessary considerations about their implementation in actual drives. For instance, the torque control is not possible with all the MTPA techniques, as well as the flux-weakening control can be easily integrated only within certain schemes. Therefore, the aim of this paper is twofold: first, to depict a clear classification strategy for the MTPA techniques, and, second, to define the most important features that an MTPA technique should implement. Last, but not least, this paper aims at summarising the proposals about the MTPA techniques, which is not present yet in the scientific literature, also reporting the MTPA problem statement in a general formal definition. The paper is expected to help future researchers in developing comprehensive MTPA techniques and practitioners in choosing among the various methods and implementations.

II. Problem statement

The torque τ produced by a synchronous motor is the following

$$\tau = \frac{3}{2} p \big(\lambda_d(i_d, i_q) i_q - \lambda_q(i_d, i_q) i_d \big) \tag{1}$$

where the variation of coenergy is neglected without loss of accuracy [9]. In (1), p is the number of pole pairs, $\lambda_d(i_d, i_q)$ and $\lambda_q(i_d, i_q)$ are the magnetic flux linkages and i_d, i_q are the dq-axes currents. In particular, the magnetic flux linkages

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are non linear functions of the currents i_d and i_q. The contemporary dependence of the magnetic flux from both currents is called *cross-saturation*. The *iron magnetic saturation* is responsible of motor nonlinearity. Both side effects sum up in the flux linkages-to-currents relationships making the current control design a quite complicated task. With the explicit flux relationships $\lambda_d(i_d, i_q) = L_d(i_d, i_q)i_d + \Lambda_{\text{mg}}(0, i_q)$ and $\lambda_q(i_d, i_q) = L_q(i_d, i_q)i_q$, where $L_d(i_d, i_q)$ and $L_q(i_d, i_q)$ are the apparent inductances, the equation (1) can be rewritten as:

$$\tau(i_d, i_q) = \frac{3}{2}p\Big[\big(\Lambda_{\text{mg}}(0, i_q) + (L_d(i_d, i_q) - L_q(i_d, i_q))i_d\big)i_q\Big] \tag{2}$$

Actually, it is often assumed that $\Lambda_{\text{mg}}(0, i_q) = \Lambda_{\text{mg}}$, i.e the permanent magnet flux linkage is considered as a constant value. It is worth highlighting that the relationship $\lambda_d(i_d, i_q) = L_d(i_d, i_q)i_d + \Lambda_{\text{mg}}$ poses the problem of defining the parameter $L_d(i_d, i_q)$ for current $i_d = 0$.

Usually, motors with predominant electromagnetic torque are called *interior permanent magnet* motors (IPM), whereas in the opposite case motors are called *permanent magnet assisted synchronous reluctance* motors (PMASynR). When only reluctance torque is present, the motor is called *synchronous reluctance* motor (SynR). It is worth to highlight that a different dq reference frame definition is normally adopted for SynR motors, where the d-axis is aligned with the direction where the reluctance at no load is at minimum. The different reference frame definition implies that $L_d(i_d, i_q) > L_q(i_d, i_q)$. This approach is not considered in the following.

There is a degree of freedom in the torque generation by the pair (i_d, i_q) that can be exploited to minimise the energy consumption. The MTPA condition can be retrieved in two ways, i.e. by minimising the current magnitude with constrained torque or by maximising the torque with constrained current magnitude:

$$\min_{(i_d, i_q)} \sqrt{i_d^2 + i_q^2} \quad \text{subject to} \quad \tau(i_d, i_q) = \tau^* \quad \text{or} \quad \max_{(i_d, i_q)} \tau(i_d, i_q) \quad \text{subject to} \quad \sqrt{i_d^2 + i_q^2} = |i| \tag{3}$$

Both ways lead to the same result, i.e. to the same MTPA condition, thus only the first of (3) is considered in the following. The choice is also motivated by the fact that the torque is imposed by the speed controller in actual application.

The solution of the first of (3) can be searched among the stationary points of the Lagrangian

$$\mathcal{L}(i_d, i_q, \ell) = \sqrt{i_d^2 + i_q^2} + \ell\big(\tau(i_d, i_q) - \tau^*\big) \tag{4}$$

i.e. the solutions of the following set of equations:

$$\frac{\partial \mathcal{L}}{\partial i_d} = \frac{i_d}{\sqrt{i_d^2 + i_q^2}} + \ell\frac{\partial \tau}{\partial i_d} = 0; \qquad \frac{\partial \mathcal{L}}{\partial i_q} = \frac{i_q}{\sqrt{i_d^2 + i_q^2}} + \ell\frac{\partial \tau}{\partial i_q} = 0; \qquad \frac{\partial \mathcal{L}}{\partial \ell} = \tau(i_d, i_q) - \tau^* = 0 \tag{5}$$

In general, it holds that $\partial\tau/\partial i_q \neq 0$. Then, solving the second of (5) for ℓ yields

$$\ell = -\frac{i_q}{\sqrt{i_d^2 + i_q^2}}\Big(\frac{\partial \tau}{\partial i_q}\Big)^{-1} \tag{6}$$

that can be replaced into the first of (5) to get the following mathematical condition for the MTPA operating point

$$i_q\frac{\partial \tau}{\partial i_d} - i_d\frac{\partial \tau}{\partial i_q} = 0 \tag{7}$$

The MTPA condition is reached when either i_d and i_q are both null (that is a trivial solution) or when the derivative terms $\partial\tau/\partial i_q$ and $\partial\tau/\partial i_d$ are both identically null. There is also a geometrical explanation of (7). One can define the gradient of the torque $\nabla\tau(i_d, i_q)$ and the current space vector in the complex domain $\mathbb{C}$. Then, the outer product of the torque gradient and the current vector is zero only when this two vectors are parallel, i.e. the MTPA condition, and the equation (7) is obtained. A similar approach is considered in [10].

The MTPA condition can be expressed even in terms of current phase angle β, see Fig. 1, i.e. considering the current vector in polar coordinates. Considering the phase angle β, it is:

$$i_d = |i|\cos(\beta) \qquad i_q = |i|\sin(\beta) \tag{8}$$

Substituting (8) into (2), for instance, the torque is function of the current magnitude $|i|$ and phase β, from which the MTPA equation can be drawn by means of the following optimisation problem:

$$\min_{\beta}(|i|) \quad \text{subject to} \quad \tau(|i|, \beta) = \tau^* \tag{9}$$

It is worth highlighting the similarities of (9) with (3). The MTPA technique expressed in polar coordinates turns out useful

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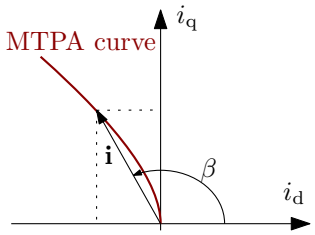


Fig. 1: MTPA curve example and definition of load angle β of current vector $\mathbf{i}$ in the dq current plane.

in several practical implementations.

Finally, it is useful to express the MTPA condition as function of one current only:

$$\frac{\partial \tau}{\partial |i|} = \frac{\partial \tau}{\partial i_d}\left(\frac{\partial |i|}{\partial i_d}\right)^{-1} = 0 \tag{10}$$

In turn, the torque equation can be rewritten as function of i_d and $|i|$ current only. The MTPA expression can be obtained by deriving the torque equation with respect to i_d and equating the results to zero.

A. MTPA formulation considering magnetic saturation

The MTPA implicit formulation can be obtained by differentiating the torque expressed as in (1) by the dq current phase angle β, which yields:

$$2l_{dq}(i_d,i_q)i_d i_q - (l_d(i_d,i_q)i_q^2 + l_q(i_d,i_q)i_d^2) + \lambda_d(i_d,i_q)i_d + \lambda_q(i_d,i_q)i_q = 0 \tag{11}$$

where l_d, l_q are the differential inductances of the d- and q-axis, respectively, and $l_{dq} = l_{qd}$ is the cross-differential inductance:

$$l_d = \frac{\partial \lambda_d}{\partial i_d} \qquad l_q = \frac{\partial \lambda_q}{\partial i_q} \qquad l_{dq} = l_{qd} = \frac{\partial \lambda_d}{\partial i_q} = \frac{\partial \lambda_q}{\partial i_d} \tag{12}$$

In order to express the MTPA condition as function of the currents only, the magnetic fluxes can be approximated as per the torque expression in (2), yielding the final MTPA expression:

$$2l_{dq}(i_d,i_q)i_d i_q - (l_d(i_d,i_q)i_q^2 + l_q(i_d,i_q)i_d^2) + L_d(i_d,i_q)i_d^2 + \Lambda_{\text{mg}} i_d + L_q(i_d,i_q)i_q^2 = 0 \tag{13}$$

It is worth noting that (13) holds for synchronous reluctance motors as well, provided that Λ_{mg} is set to zero. The MTPA formulation in (13) is seldom used due to the difficulties about the precise knowledge of the differential inductances. Future research activities should consider the effect of the differential inductances on the MTPA point determination. Recent examples of the formulation (13) can be found in [10], [11].

B. MTPA formulation with constant parameters

In practice, equation (2) is often used. By applying (10) on (2), the following equation with constant parameters is obtained:

$$i_d = \frac{\Lambda_{\text{mg}} \pm \sqrt{\Lambda_{\text{mg}}^2 + 8(L_d - L_q)^2|i|^2}}{-\Lambda_{\text{mg}}(L_d - L_q)} \tag{14}$$

Equation (14) can be obtained from (13) by imposing $l_{dq} = 0$, $l_d(i_d,i_q) = L_d$, $l_q(i_d,i_q) = L_q$ and solving by i_d. In other words, the result of (14) is the correct MTPA point only if the motor magnetic flux characteristics are linear and no cross-saturation is present. Quite often the equation (14) is implemented by adapting the motor parameters value based on the working point, trying to get closer to the actual MTPA point. In turn, L_d and L_q in (14) are function of both i_d and i_q. However, the result is still not the correct MTPA point, since only apparent inductances are considered and the magnetic cross-saturation is not accounted in (14).

For the sake of generality, the MTPA problem statement was also studied in a very accurate formal way in [8].

III. Classification

In order to outline a classification strategy, it is important to highlight the strong dependency of the MTPA condition on motor parameters. Therefore, classification of MTPA techniques can be carried out according to the method adopted for parameters extraction. However, some techniques may not require the knowledge of the motor parameters. Therefore, a more general classification approach is to consider if the MTPA curve has been obtained prior the motor is installed in the final application.

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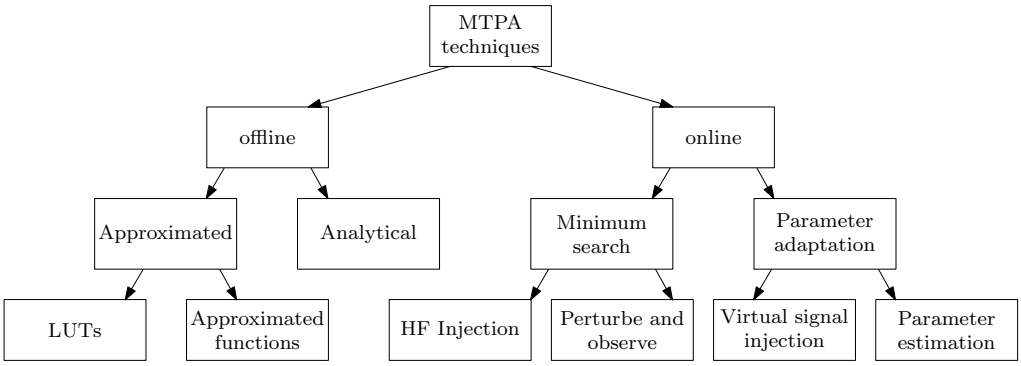


Fig. 2: Classification tree of the MTPA techniques.

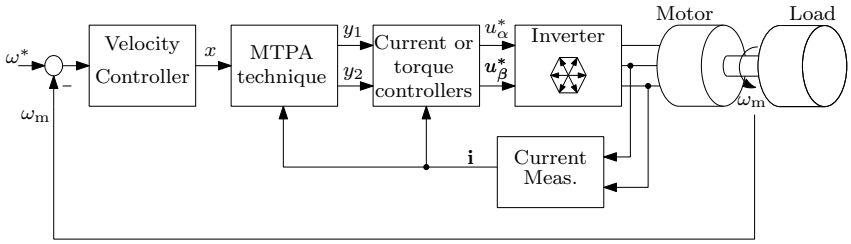


Fig. 3: General electric drive control scheme with MTPA techniques.

Bearing this in mind, the MTPA techniques can be classified in two main groups: *offline* and *online* techniques. The former are discussed in Sect. IV, whereas the latter are in Sect. V. Using this general classification, the tree-diagram of Fig. 2 is obtained.

The branches of the tree in Fig. 2 are discussed in the rest of the paper. However, there are other characteristics which can be part of the classification. For instance, *model-free* or *model-based* techniques, i.e. the need of motor parameters knowledge, are both possible in offline and online MTPA based controllers. Some techniques adopt an offline strategy at the very beginning of the motor drive life, and then a second algorithm is used to update the MTPA curve. This like of algorithms are referred to as *hybrid algorithms*.

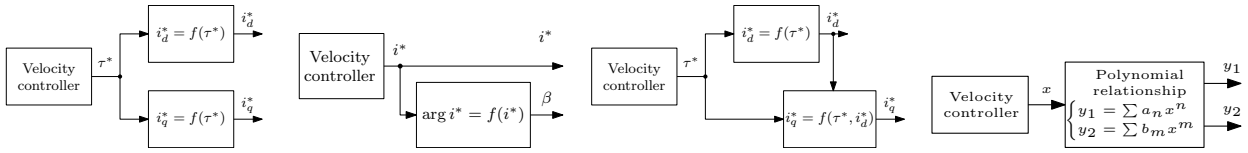
A. Implementation schemes

The MTPA techniques are usually implemented by processing the output of the speed regulator producing a set of references for the torque or current control loops. A general schematic of electric drives performing MTPA operations is sketched in Fig. 3. Actually, the input x and the outputs (y_1, y_2) of the MTPA block in the latter figure can be different based on the MTPA technique. On one hand, *direct torque control* (DTC) techniques do not require current references, but the inputs are still the torque reference from the speed controller and a flux reference value that has to be somehow obtained, as it will be discussed in Sect. IV-E. On the other hand, *Field Oriented Control* (FOC) based techniques usually consider the dq current references as outputs, i.e. i_d^* and i_q^* or their equivalent polar representation, while the input is usually represented by the speed regulator output (reference value for current magnitude or torque).

Several possibilities for MTPA implementation within FOC are reported in Fig. 4. The techniques can be either of the online or offline type. In the latter case, there is no current feedback employed in the MTPA technique as reported in Fig. 3. In some meaning, they appear to be as *open-loop* techniques and there is no feedback action to correct their possible misbehaviour. Apparently, hybrid solutions of online and offline techniques represent still an open research field, although some implementations have been already proposed [12], [13]. It is worth reporting that some proposals in technical papers implement variations of schemes in Fig. 4 (such as in [14]), in particular when presenting wide speed range drives including *field weakening* (FW) or *maximum torque per voltage* MTPV. Furthermore, techniques aiming at minimising the power consumption can take into consideration the mechanical speed as an input, too. Techniques based on polynomial functions and analytical solutions of (3) present structures like the one in Fig. 4d. The actual calculation blocks are replaced by approximating functions, which enable the MTPA-based control of the drive.

Online techniques rely on feedback information to work out the MTPA condition. The motor current measurements are thus needed, but, in some cases, additional measurements are also required, e.g. voltage or speed. In fact, the motor speed measurement oscillations are sometimes exploited for tracking the MTPA condition. In other cases, voltage and current values are processed in order to estimate motor parameters or to evaluate variations of active power. The block schemes of Fig. 4 fit

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(a) Torque reference as input (b) Current reference as input (c) Towards a FW implementation (d) Polynomial based implementation

Fig. 4: Different solutions for implementing MTPA techniques.

for the majority of online MTPA techniques. Of course, measurements are needed, as reported in the sketch of Fig. 3, while in some cases the function blocks may be replaced by tracking algorithms.

IV. Offline methods

A. Look-Up Table based solutions

Techniques belonging to this categories are the most common in the scientific literature. Taking the schemes of Fig. 4 as the reference, function blocks represented there are often implemented (in an approximated way) by means of Look-Up Tables (LUTs). In the following, variant solutions will also be highlighted and discussed. In turn, the differences among the solutions in Fig. 4 stand in the choice of the input for the LUTs. Transient operations of three LUTs based MTPA techniques are compared in [14]. The scheme in Fig. 4b is considered and modified, showing that better performances could obtained when current measurements are used as LUT input instead of the reference outgoing from the speed controller. In more complicated schemes aiming at wide speed range control, an additional input may be required. The scheme in Fig. 4a was implemented in [15] for validating the experimental identification of the magnetic model of the motor. The choice of the torque reference τ^* as input is related to the availability of the complete relationships between flux linkages and currents.

Actually, the scheme in Fig. 4a seems to be the preferred one in wide speed range drives. Several proposals have been presented in this field, such as [16]–[19]. The proposal in [16] extends the speed range by an additional input to the LUTs, i.e. the desired magnetic flux, to the torque reference. The control scheme is similar to Fig. 4a, but with two inputs for each LUT. The work in [17] presents the same MTPA implementation structure as in [16]. The LUTs were obtained by means of a laboratory test bench equipped with wattmeter and torquemeter. A different way was chosen in [18], still similar in the principle of operation to the scheme in Fig. 4a. In fact, the LUTs are used to obtain the exact flux linkages which in turn are used to compute the optimal current references for minimum copper losses.

Open questions about the LUT implementation are still present, in particular regarding the optimal number of points to be used (and their distribution in the range) and the management of out-of-range inputs. The former affects the accuracy of the MTPA approximation and the computational burden. A possible solution is to set a number of maximum points and place them in such a way to minimise the variation of slope change after each point or maximizing the accuracy of MTPA. The latter aspect regards the possible overload operations to be carried out for a short amount of time. In these transients, it is mandatory to avoid the production of unfeasible currents due to poor extrapolation algorithms. A possible solution is to adopt the slope defined by the last two elements of the LUT and assume it will stand for any inputs bigger than the last entry of the LUT. Since the MTPA curve usually reaches a slant asymptote, the schemes based on polar coordinates seem to offer good extrapolation accuracy by simply keeping the extreme angle value for out-of-range current magnitude.

B. Analytical solutions

The analytical solution of (14) is often used for achieving the MTPA condition. The motor parameters are often assumed to be constant. The knowledge of the magnetic flux relationship with the currents can be exploited to calculate (14), usually considering variable inductances. Such approach is adopted in [20]–[23]. In [20], [21], the MTPA current references are combined with a flux weakening algorithm to widen the speed region of the drive. This is one of the first adopted techniques, due to the simplicity of its method. However, it brings along an error in the MTPA point determination as reported in Sect. II-B.

C. Polynomial solutions

Various approaches can be found for MTPA techniques based on polynomial functions. The method proposed in [24] uses the computation of power losses to estimate the difference between L_d and L_q, which is in turn exploited to compute the current references for the MTPA condition. The strategy is based on electrical and mechanical constant parameters. The minimisation of the power consumption is the goal of [25], too. Simple relationships of the magnetic flux with respect to the currents and constant parameters for approximating the iron losses are employed.

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The MTPA equation (14) is also expanded by means of the Taylor' series. Equation (14) is thus expressed by polynomials which yield the dq current references, such as:

$$i_d^*(x) = \sum_{n=0}^{N} a_n x^i \qquad i_q^*(x) = \sum_{m=0}^{M} b_m x^i \tag{15}$$

The number of coefficients is kept as low as possible for the sake of less computational effort and for avoiding the well-known oscillatory behavior of high-degree polynomial fitting. For instance, two simple equations are used in [26], where $i_q^* = f(\tau^*)$ and $i_d^* = f(i_q^*)$. From the representation point of view, the schematic of the control proposed in [26] is identical to Fig. 4b. Also [27]–[29] uses the Taylor' series approximation of (14). The former proposes an MTPA control for a sliding-mode based sensorless drive. Constant parameters are used in the latter algorithm, which makes it an offline algorithm from the MTPA point of view. The approximated MTPA relationships are applied in [29] to a neural network based speed controller.

Equation (14) is also rearranged in different forms. In order to take advantage of the estimated PM flux linkage, [30] proposes a formulation of (14) that is also suitable for online computation. A linear relationship with the speed controller output is obtained in [31], which also proposed a wide speed range drive.

D. Power losses minimisation

It is worth mentioning the motivations behind the choice between MTPA and Minimum Loss Algorithm, i.e. converter vs motor ratings. The problem of power losses minimisation has been investigated since the '80s [1]–[3], [32], when the first FOC drives were implemented. First attempts to develop a loss minimisation control are often based on constant parameters, such as in [33]. The methods proposed in [34]–[38] resemble the analytical calculation of the MTPA condition with constant parameters in (14). Usually, the partial derivative of the power respect to the d-axis current, i.e. $\frac{\partial P}{\partial i_d} = 0$, is reported in those articles. Differences among the various methods are mostly in the way iron losses are calculated. In order to get a viable solution, dedicated experiments with or without additional equipment have to be carried out, or heavy assumptions are made. Another approach is through FEA results, which is only possible when geometrical and material data of the motor are available. Moreover, accurate simulation of iron loss effects is not straightforward [39], [40]. For such reasons, the offline approach to attain MTPA seems simplistic, compared to the complexity of the problem. Future research may improve power loss minimization methods with online adjustments, with the aim of attaining actual maximum efficiency operation.

In [34], the copper, iron and additional losses are the quantities object of the minimisation in place of the torque at equation (14). Analytical models of the latter losses are adopted at first, but an experimental procedure based on a three-phase wattmeter measurements is also proposed. In [35], the resistance variation is considered by means of Fourier's equation, while the iron losses are calculated by taking advantage from Epstein measurements combined with FEA results on the motor under test. The bearings losses model is obtained from datasheet of the mechanical component. Also [41] takes advantage from FEA results to develop a maximum efficiency algorithm. The iron losses are calculated from dedicated experiments in [37], [38], while the inductance L_q is approximated by a linear function of the current i_q. Still FEA results allowed the authors in [42] to solve a fourth degree problem with Ferrari's solution.

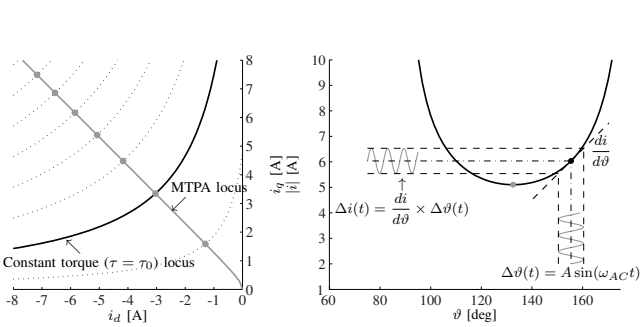
E. Direct Torque Control

Since the torque is due to the interaction between magnetic flux and currents, it is possible to generate the desired torque by controlling the product of magnetic flux by the stator currents. Normally, the magnetic flux is expressed in terms of stator currents and the torque is controlled by means of current references. The basic principle of a DTC scheme is that the magnetic flux vector amplitude can be controlled directly from the applied voltage. Thus, the current control is avoided and a reliable torque production is obtained by stating a proper flux-linkage angle [43], [44]. However, the MTPA paradigm has been only recently investigated for DTC applications. It is worth reporting here that other methods exist, such as the *direct flux field-oriented control*, that resemble the form of a DTC, but include PI control loops and Pulse Width Modulation. An example is [45], where the MTPA curve was obtained previous the motor starts operations. A similar control scheme is also proposed in [46] for a SynR motor.

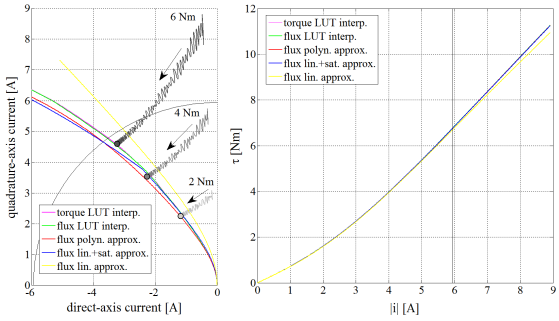
Most of the techniques adopting DTC uses the model of the motor to get the MTPA condition. A polynomial approximation of the MTPA curve is used in [47], which requires the knowledge of at least three MTPA points prior the motor starts nominal operations. Therefore, this technique is classified as an offline one. Similar proposals to [47] are [48], [49].

A parameter estimation based method is proposed in [50]. It relies on the inductance estimation during the motor operations, which allows to consider [50] as an online technique. An example of online technique is reported in [51], while on-line or hybrid implementation is proposed in [52]. It is one of the few examples of MTPA technique employed in a DTC based scheme that does not require motor parameters, but it resorts to the injection of signals to work out the MTPA condition.

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(a) Small-signal approach for MTPA tracking according to the low-frequency injection method [53]



(b) Different MTPA trajectories, based on torque or flux measurements or flux maps approximations.

Fig. 5: MTPA online techniques

V. Online methods

Parameters uncertainty and variability led to the idea of on-line or adaptive MTPA methods, i.e. algorithms aiming at identifying and/or refining the MTPA operating point, exploiting information obtained during normal operation of the drive.

In methods based on adaptation ("model-based"), e.g. [54]–[60], current references are calculated using formulas (e.g. (14)), but motor parameters (L_d, L_q, Λ_{mg}) are estimated on-line using different approaches, ranging from closed-loop schemes [54] to recursive least squares [58], [59]. Such methods are complex and cross-saturation is usually not considered. Other methods numerically solve the MTPA problem using flux maps updated on-line, e.g. using Artificial Neural Networks [60].

Signal injection approaches to on-line MTPA search, called MTPA tracking [51], [53], [61]–[68], have been proposed in the last decade. A small sinusoidal perturbation is applied to the current vector reference, while the operating point is adapted based on machine response to such stimulus. The basic idea is to demodulate a measured or estimated variable, which depends on the perturbation. The resulting signal is an out-of-MTPA indicator, i.e. it crosses zero in the MTPA point. A closed-loop regulator (typically PI) controls that error signal to zero, thus tracking the MTPA point. In this way, accuracy of MTPA does no more depend on machine parameters, differently from other approaches.

Two types of injection have been considered. In the first one, [51], [61]–[65], a small signal at relatively high frequency (e.g. hundreds of Hz) is added to the current vector angle. Current vector magnitude is virtually not affected, but torque oscillation is detected by processing measured speed or estimated active power, i.e. the problem is stated as in the second of (3). A different operating principle is applied in the low-frequency methods [53], [66]–[68]. In this case, current signal injection is at lower frequency, ideally within the speed control bandwidth (e.g. tens of Hz). Speed regulation forces constant torque (i.e. problem as stated in (3)), while the current vector magnitude oscillates, as shown in Fig. 5a. These methods rely on load torque being constant or slowly-varying, thus in some cases they are enabled only at steady-state [63], although robustness to moderate load transients has been demonstrated [67]. Since these techniques imply feedback, stability and convergence need consideration, as shown in recent papers [53], [66]–[68]. Fig. 5b shows an example of MTPA tracking traces (grey), compared to MTPA curves obtained offline, based on various methods, namely torque measurements, flux maps and flux approximations (polynomial, linear+saturation [69] and constant inductance) for an IPMSM. Torque vs. current is proposed as an effectiveness test for MTPA methods, also considering different sources of motor parameter data (e.g. torque measurements, flux measurements, self-identification of magnetic parameters).

Other minimum-search techniques are based on slow variation of the current vector angle [70] or "perturbate & observe" (P&O) [56], [71], [72]. The main drawbacks of such methods are related to their dynamics and robustness to load variations, which are addressed in [56], where P&O is combined with parameters estimation. Recently, Virtual Signal Injection Control (VSIC) has been proposed [13], [73], [74], aiming at MTPA tracking without injection. This class of techniques is actually based on estimation of variables rather than their derivative, and its accuracy has been questioned [75].

VI. Conclusions

Research production on MTPA techniques is massive, with many proposals and claims, which could be confusing for a reader. In this paper, MTPA methods have been reviewed and classified considering their operating principles, implementation and limitations. The common problem statement underlying every MTPA method has been stated and commented, clarifying the common aim and listing most of the variants proposed in literature. A comprehensive classification has been carried out, analysing several aspects, such as the off- or on-line nature of the method, the source of motor data and the way in which that is exploited.

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The paper summarises the state-of-the-art on MTPA techniques, gathering useful information for guiding researchers and practitioners in the development of novel ideas or implementation of known methods, while highlighting those aspects which would benefit from further investigation.

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